formance in the workplace, manage stress, and also help them adopt healthier and more productive working styles.

• Voice-based emotion analysis in real time opens up many business opportunities by enabling an automated customer service agent to recognise a caller's emotional state and adapt accordingly. Such information can also be helpful in analysing and managing stress levels of human workers.

• Emotion-sensing technology will also enable companies to establish deeper emotional connections with their consumers through virtual assistants. Data collected through such devices can help companies understand how internal and external environmental factors impact their employees. As a result, companies can redesign processes accordingly, to help keep their personnel better engaged and productive.

• Emotion sensing through wearables can help monitor the user's state of mind in terms of mental and other health conditions.

Mood sensor technology can help children or the elderly, who are in need of care, to receive timely assistance and support from family members or caregivers, who can then alert doctors.

• Technology that deduces human emotions based on audio-visual cues may enable businesses to detect consumers' positive and negative moods to better understand their preferences. Such technology can help analyse customer choices that can be utilised for marketing and help detect their annoyances to improve product usability.

• Emotion-sensing smart home devices could provide entertainment (music, videos, TV shows or imagery) to match the user's current state of mind.

A fridge with a built-in emotion sensor may interpret a person's mood and suggest a suitable food item for the user to consume.

Similarly, video games can use emotion-based biofeedback technology to adjust game and difficulty levels according to the player's emotional state.

In the future

Various fields including medicine, advertising, robotics, virtual reality, gaming, education, working conditions and safety, automotive, home appliances and so on will significantly benefit from the use of emotion-sensing technology. Newly-developed smart devices will be able to arrest emotional reactions and convert these into data and facts to analyse situations and come up with appropriate recommendations accordingly.

At present, healthcare and automotive industries are among the most eager to adopt emotion-sensing features. Car manufacturers are exploring the implementation of in-car emotion-detection systems to improve road safety by managing the driver's drowsiness, irritation and anxiety.

Owing to achievements in computer vision, the emotion-sensing technology field is progressing on understanding human emotions, speech recognition, deep learning and related technologies. Every year, we see new mood sensor technologies being realised. While most of the existing emotion-sensing inventions are based on the use of on-body devices or voice/facial recognition software, research and development efforts are increasingly being directed towards sensing technology that can measure emotions in a contactless manner. EFY

SenseLabs' headset trains mental athletic performance (Credit: www.sporttechie.com)